

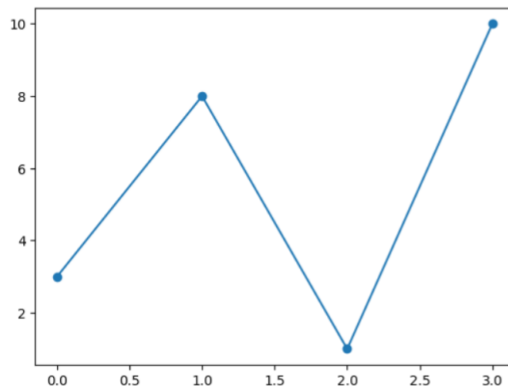
MATPLOTLIB

Program 1: Basic Line Plot with Marker

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])  
plt.plot(ypoints, marker='o')  
plt.show()
```

```
[1]: import matplotlib.pyplot as plt  
import numpy as np  
  
ypoints = np.array([3, 8, 1, 10])  
  
plt.plot(ypoints, marker='o')  
plt.show()
```



Program 2: Star Marker

```
import matplotlib.pyplot as plt  
import numpy as np
```

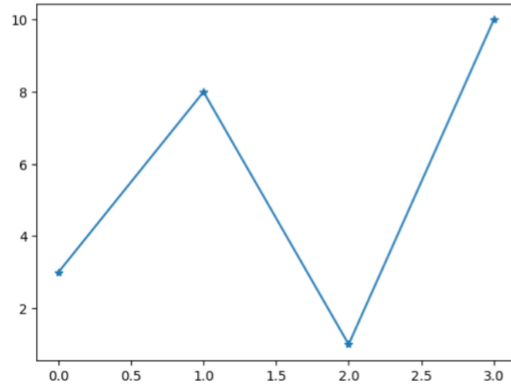
```
ypoints = np.array([3, 8, 1, 10])  
plt.plot(ypoints, marker='*')  
plt.show()
```

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```
[2]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker='*')
plt.show()
```



Program 3: Format String (marker + line + color)

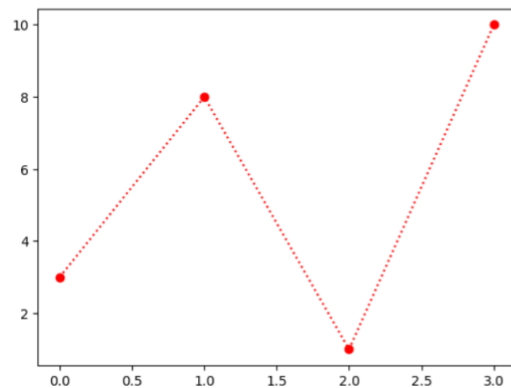
```
import matplotlib.pyplot as plt
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
plt.plot(ypoints, 'o:r')
plt.show()
```

```
[3]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, 'o:r')
plt.show()
```



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Program 4: Marker Size

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
```

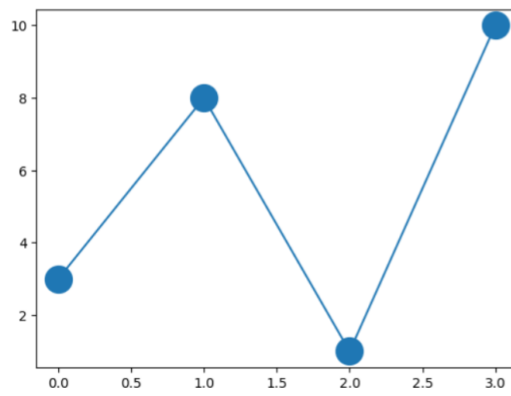
```
plt.plot(ypoints, marker='o', ms=20)
```

```
plt.show()
```

```
[4]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, markers='o', ms=20)
plt.show()
```



Program 5: Marker Edge Color

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
```

```
plt.plot(ypoints, marker='o', ms=20, mec='r')
```

```
plt.show()
```

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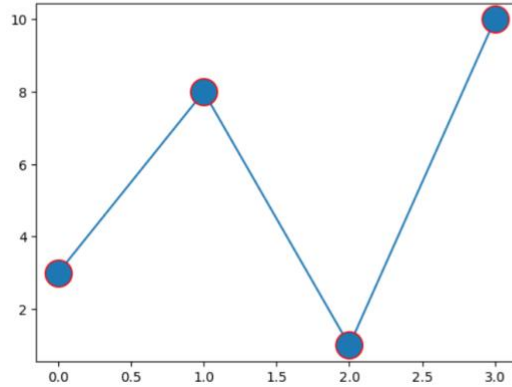
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```
[5]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker='o', ms=20, mec='r')
plt.show()
```



Program 6: Marker Face Color

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
```

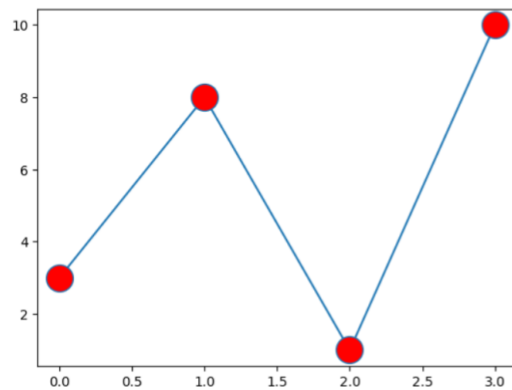
```
plt.plot(ypoints, marker='o', ms=20, mfc='r')
```

```
plt.show()
```

```
[6]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker='o', ms=20, mfc='r')
plt.show()
```



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Program 7: Marker Edge and Face Color

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
```

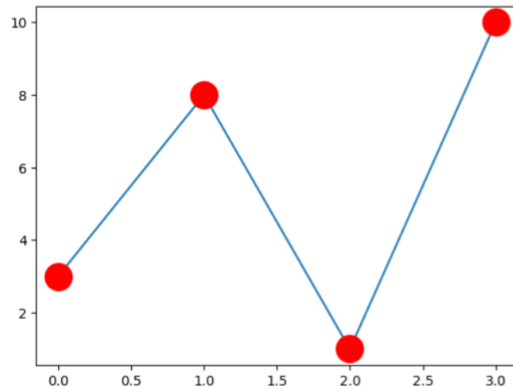
```
plt.plot(ypoints, marker='o', ms=20, mec='r', mfc='r')
```

```
plt.show()
```

```
[7]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, marker='o', ms=20, mec='r', mfc='r')
plt.show()
```



Program 8: Dotted Line

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
```

```
plt.plot(ypoints, linestyle='dotted')
```

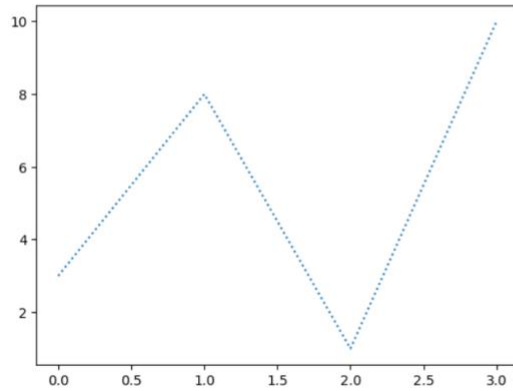
```
plt.show()
```

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```
[8]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, linestyle='dotted')
plt.show()
```



Program 9: Dashed Line (Short Syntax)

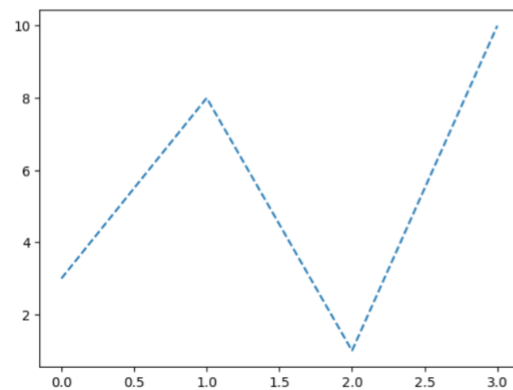
```
import matplotlib.pyplot as plt
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
plt.plot(ypoints, ls='--')
plt.show()
```

```
[9]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, ls='--')
plt.show()
```



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Program 10: Line Color

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
```

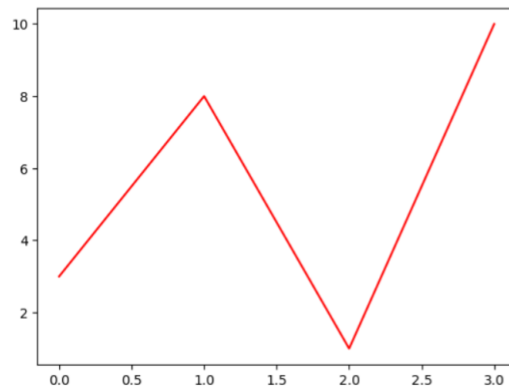
```
plt.plot(ypoints, color='r')
```

```
plt.show()
```

```
[10]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, color='r')
plt.show()
```



Program 11: Line Width

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
ypoints = np.array([3, 8, 1, 10])
```

```
plt.plot(ypoints, linewidth=5)
```

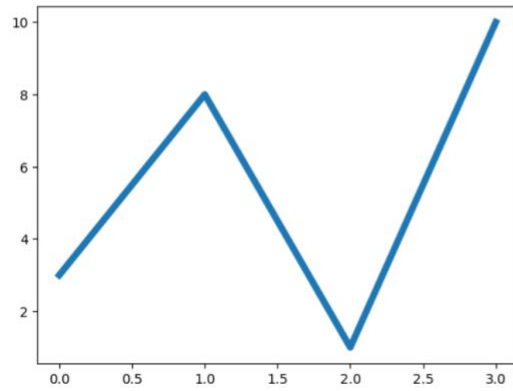
```
plt.show()
```

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```
[11]: import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, linewidth=5)
plt.show()
```



Program 12: Multiple Lines (Separate Calls)

```
import matplotlib.pyplot as plt
import numpy as np
```

```
y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])
```

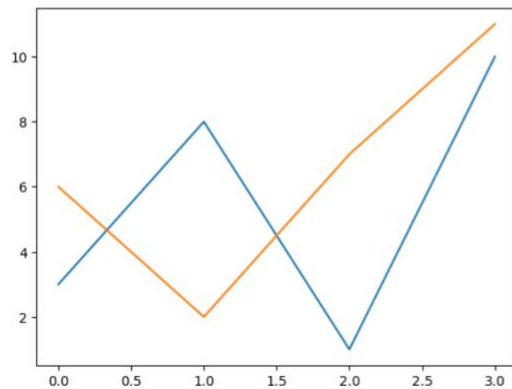
```
plt.plot(y1)
plt.plot(y2)
plt.show()
```

```
[12]: import matplotlib.pyplot as plt
import numpy as np

y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])

plt.plot(y1)
plt.plot(y2)

plt.show()
```



Program 13: Multiple Lines with X and Y Values

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x1 = np.array([0, 1, 2, 3])  
y1 = np.array([3, 8, 1, 10])
```

```
x2 = np.array([0, 1, 2, 3])  
y2 = np.array([6, 2, 7, 11])
```

```
plt.plot(x1, y1)  
plt.plot(x2, y2)  
plt.show()
```

```
[13]: import matplotlib.pyplot as plt  
import numpy as np  
  
x1 = np.array([0, 1, 2, 3])  
y1 = np.array([3, 8, 1, 10])  
  
x2 = np.array([0, 1, 2, 3])  
y2 = np.array([6, 2, 7, 11])  
  
plt.plot(x1, y1)  
plt.plot(x2, y2)  
  
plt.show()
```

